

Ph.D. Qualifying Exam 2018: Statistical Inference

August 22, 2018
10 AM to 2 PM

Rules: This is a four hour exam. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “cheat sheet”. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Good luck!

Problem 1

Name _____

We observe the single discrete random variable X having probability mass function

$$f_{\theta}(x) = P_{\theta}(X = x),$$

where

$$f_{\theta}(x) = \begin{cases} \theta/6, & x = 0, \\ \theta/2, & x = 1, \\ 1 - \theta, & x = 2, \\ \theta/3, & x = 3. \end{cases}$$

The parameter space is $\Theta = \{\theta : 0 \leq \theta \leq 1\}$.

- (a) Does this family have the monotone likelihood ratio property in x ? (Prove or disprove)
- (b) We want to test the null hypothesis $H_0 : \theta \leq 1/4$ versus the alternative $H_1 : \theta > 1/4$.
 - (i) Does there exist a UMP level α test? Prove or disprove. If so, find it and establish that it is a UMP level α test.
 - (ii) Using the Uniform distribution on $[0, 1]$ as a prior distribution for θ what is the size of the Bayes test procedure?
- (c) Does a complete sufficient statistic exist? If so, find it and prove that it is complete sufficient.
- (d) Does there exist a uniformly minimum variance unbiased estimator (UMVUE) of θ ? (Prove or disprove). If so, obtain it and establish that it is a UMVUE estimator.

Problem 2

Name _____

Let $\{(X_i, Y_i) : i = 1, \dots, n\}$ be an iid sample, where Y_i is a binary variable taking values in $\{0, 1\}$ and X_i is a $p \times 1$ vector of covariates. Consider a logistic model

$$P(Y_i = 1|X_i) = \pi(X_i; \beta),$$

where β is a vector of unknown parameters and

$$\pi(X_i; \beta) = \frac{e^{X_i^T \beta}}{1 + e^{X_i^T \beta}}.$$

Assume that the model is correctly specified, and denote by β^* the true coefficient vector. Let $\hat{\beta}$ be the maximum likelihood estimator (MLE) of β . You may use the result that under suitable conditions, $\sqrt{n}(\hat{\beta} - \beta^*)$ is asymptotically distributed as $N(0, U_{\beta^*})$ as $n \rightarrow \infty$ and p is fixed, where U_{β^*} is a variance matrix.

(i) Let $\tilde{\beta}$ be an estimator defined as a solution to the equation,

$$\frac{1}{n} \sum_{i=1}^n \frac{Y_i - \pi(X_i; \beta)}{\sqrt{\pi(X_i; \beta)(1 - \pi(X_i; \beta))}} X_i = 0.$$

Show that there exists a loss function $\kappa(\beta)$ such that

$$\tilde{\beta} = \operatorname{argmin}_{\beta} \kappa(\beta),$$

and express $\kappa(\beta)$ in as simple a form as possible.

(ii) For fixed p , show that $\tilde{\beta}$ is a consistent estimator of β^* as $n \rightarrow \infty$. You may impose technical conditions as needed, but weaker conditions are preferred.

(iii) For fixed p , show that $\sqrt{n}(\tilde{\beta} - \beta^*)$ is asymptotically distributed as $N(0, V_{\beta^*})$ as $n \rightarrow \infty$, where V_{β^*} is a variance matrix. You may impose technical conditions as needed, but weaker conditions are preferred.

(iv) Determine V_{β^*} in (iii) in as simple a form as possible and show that $V_{\beta^*} \geq U_{\beta^*}$ in the sense that the difference $V_{\beta^*} - U_{\beta^*}$ is non-negative definite. Recall that U_{β^*} is the asymptotic variance of the MLE $\hat{\beta}$. Is this result intuitive? Explain why or why not in a sentence or two.